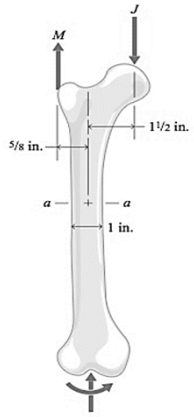
Internal Reactions

Example:

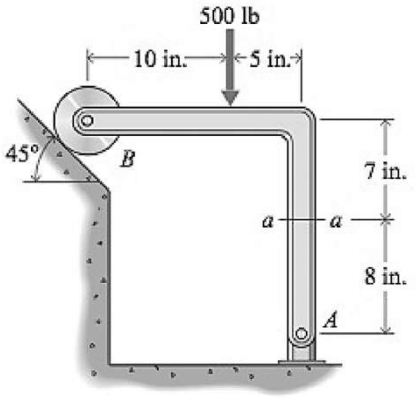
A human femur is modeled in the figure below. The abductor muscle force is M = 90 lb, and the femoral load is J = 120 lb. Determine the internal forces on section a-a.



Normal internal reaction force = 30 lb (C)
Shear reaction force = 0 lb
Moment (about the cut) reaction = 236 lb.in

Example:

Determine the internal forces on section a-a in the angle bracket shown in the Figure below.

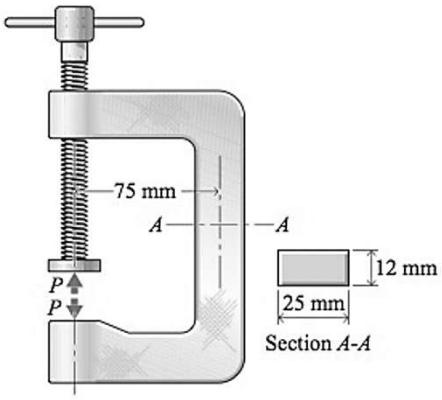


Normal internal reaction force = 417 lb(C)
Shear internal reaction Force = 83.3
Moment = 236 lb.in (CW)

Internal Reactions

Example:

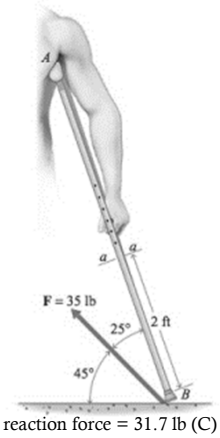
Determine the internal forces acting at the centroid of section A-A of the C-clamp shown in the figure below. The force P has a magnitude of 2000 N.



Normal internal reaction force = 2000 N
Shear reaction force = 0 lb
Moment (about the cut) reaction = 150 N.m

Example:

The reaction between a crutch and the ground is 35 lb, as shown. Determine the internal forces acting at section a-a.

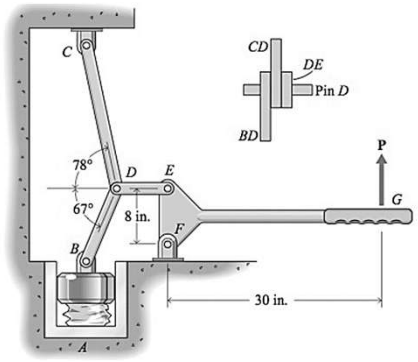


Normal internal reaction force = 31.7 lb (C)
Shear reaction force = 14.79
Moment (about the cut) reaction = 29.6 lb.ft (CCW)

Internal Reactions

Example:

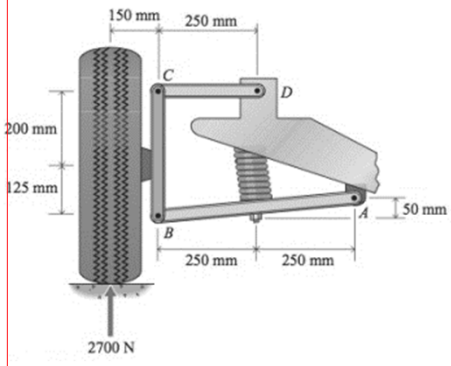
A pin-connected system of levers and bars is used as a toggle for a press as shown below. Three members are joined by pin D, as shown in the insert. Determine the internal forces on a cut perpendicular to and midway between the 2 force member DE when P = 1000 lb.



Normal internal reaction force = 3750 lb
Shear reaction force = 0
Moment (about the cut) reaction = 0

Example:

The front-wheel suspension of an automobile is shown below. The pavement exerts a vertical force of 2700 N on the tire. Determine the internal forces on a surface perpendicular to the horizontal and midway between C and D. (Hint: Member CD is a 2-force member)

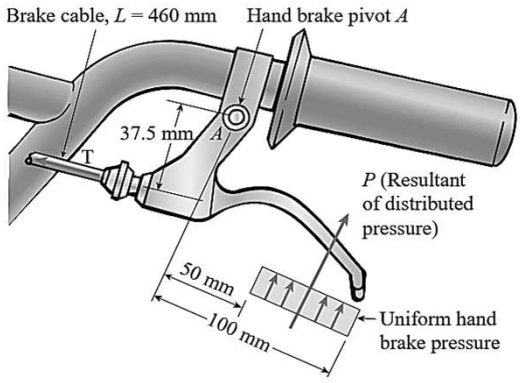


Normal internal reaction force = 1246 N(C)
Shear internal reaction Force = 0
Moment = 0

Internal Reactions

Example:

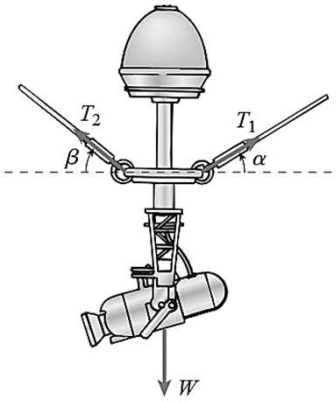
A force P of 70 N is applied by a rider to the front hand brake of a bicycle (P is the resultant of an evenly distributed pressure). As the hand brake pivots at A, a tension T develops in the 460-mm long brake cable. Find the internal reactions in the brake cable.



Normal internal reaction force = 140 N
Shear reaction force = 0
Moment (about the cut) reaction = 0

Example:

Two steel wires support a moveable overhead camera weighing 25 N (see figure below) used for close-up viewing of field action at sporting events. At some instant, wire 1 is at an angle $\alpha = 20°$ to the horizontal and wire 2 is at an angle $\beta = 48°$. Determine the internal reactions in the two wires.



Normal internal reaction forces (Wire 1, Wire 2) = 25.34 N, 18.04 N.